Roll No.

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4E1301

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B. Tech. IV - Sem. (Main / Back) Exam., - 2025
Computer Sc. & Engineering (Cyber Security)
4CCS2-01 Discrete Mathematics Structure
CS, IT, AID, CAI, CCS, CDS, CIT

Time: 3 Hours **Maximum Marks: 70**

Instructions to Candidates:

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART – A

[10×2=20]

(Answer should be given up to 25 words only)

All questions are compulsory

Q.1 If A = {1, 2, 3} and B = {a, b, c}, how many total possible relations can be formed from A to B?

Q.2 If f, g : $\mathbb{R} \rightarrow \mathbb{R}$ are defined as $f(x) = x^2$ and g(x) 5x + 1. Find the composition $(f \circ g)(x)$ at x = 2.

Q.3 When are two finite state automata said to be equivalent?

Q.4 Define universal and existential quantifiers.

Q.5 What is a complemented lattice?

Q.6 Find the middle term in the expansion of $(x^2 + 2/x^4)^{14}$.

Q.7 What is an Abelian Group? Give an example of an infinite Abelian group.

Q.8 Define the cyclic group.

Q.9 What do you mean by a regular graph?

Q.10 Draw graph which is -

(a) Eulerian but not Hamiltonian.

(b) Hamiltonian but not Eulerian.

PART – B

[5×4=20]

(Analytical/ Problem solving questions)

Attempt any five questions

Q.1 Among 50 students in a class, 26 got an A in the first examination and 21 got an A in the second examination. If 17 students did not get an A in either examination, how many students got A in both the examinations?

Q.2 Use mathematical induction to prove that for all $n \geq 1$, $10^n - 1$ is divisible by 9.

Q.3 Prove by constructing truth table

$$P \rightarrow (Q \vee R) \cong (P \rightarrow Q) \vee (P \rightarrow R).$$

Q.4 Let B be the power set of $S = \{1, 2, 3\}$ and $(B, \leq)$ be a poset defined by $X \leq Y$ if $X \subseteq Y$ for $X, Y \in B$. Draw the Hasse diagram of the poset $(B, \leq)$.

Q.5 How many words can be formed from the letter of the word "DAUGHTER" if the vowels always coming together?

Q.6 Prove that the intersection of any two normal subgroups of a group G is normal subgroup of G.

Q.7 Determine whether the graph given below by its adjacency matrix is connected or not.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

PART – C

[3×10=30]

(Descriptive/ Analytical/ Problem Solving/ Design Questions)

Attempt any three questions

Q.1 Let R be a relation defined on the set $\mathbb{N} = \{1, 2, 3 \ldots\ldots\ldots\}$ such that a, b ∈ ℕ, aRb if and only if $a = b^k$, $k \in (0, 1, 2, 3, \ldots\ldots)$. Then show that R is a partial ordering relation on ℕ.

Q.2 (a) Find the disjunctive normal form of

$$P \rightarrow ((P \rightarrow Q) \wedge \sim(\sim P \vee \sim P)).$$

(b) Show that the following proposition is tautology:

$$(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)).$$

Q.3 Find an explicit formula for the following linear homogeneous recurrence relation :

$$a_n = -4a_{n-1} - 3a_{n-2}, \text{ if } n \geq 2,$$

with the initial conditions $a_0 = 4$ and $a_1 = 8$.

Q.4 Show that the set of all square matrix of order (m × m) under the binary operations addition and multiplication is a non-commutative ring.

Q.5 Using Dijkstra's algorithm, find the shortest path between the vertices A and H in the weighted graph shown in figure.